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Layout-to-Image Driven Data Augmentation for Object Detection

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***Abstract*—Supervised object detection models require sufficient annotated datasets to achieve high accuracy. However, collecting and labelling large-scale datasets with accurate bounding boxes is labor-intensive. In this paper, we propose a novel data augmentation strategy that leverages diffusion-based generative models conditioned on bounding box layouts. By adopting the layout-to-image model, large-scale, realistic images can be automatically generated with the input of object locations and category labels. The ground truth label and image pairs can be obtained without manual labeling. A comprehensive experiment has been conducted to evaluate the effectiveness of the generated images and labels. Experimental results demonstrate that adopting the proposed layout-conditioned diffusion method can effectively reduce annotation workload and enhance detector robustness. To be more specific, using 10,000 generated images and label pairs can improve 3.12% mean Average Precision on the COCO dataset, and 3.14% on the PASCAL VOC dataset.**

Keywords—Object detection, diffusion models, image generation, layout to image (L2I), data augmentation

I. Introduction

Object detection is one of the most famous perception tasks. Deep learning-based object detection models have achieved significant success in recent years. However, their performance is highly dependent on the quantity and diversity of training data [1], [2], [3], [4]. Acquiring large-scale, richly annotated image datasets is challenging because each image requires careful placement of bounding box annotations around the objects. Insufficient data or limited diversity may lead to overfitting or failure to generalize to new scenarios [5], [6]. Therefore, data augmentation has become an essential technique for expanding and diversifying training datasets without requiring additional manual annotation. Traditional augmentation strategies (such as random flipping, scaling, rotation, and mosaicking) generate modified images that retain the original content and bounding box labels to enhance model robustness. However, these methods only introduce minor perturbations to existing images, resulting in limited diversity in newly introduced training samples [7-10]. More complex augmentation methods attempt to insert new object instances or mixed scenes (e.g., copy-paste methods). However, they often struggle to maintain semantic consistency within the scene, which may confuse the detector rather than aiding its recognition [10], [11]. These challenges highlight the need for augmentation techniques that simultaneously diversify the dataset while preserving the structure of real-world scenes.

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Generative approaches provide a new direction for data augmentation by creating entirely new images [10], [12]. Recent developments in deep generative models – in particular, diffusion models – have created highly realistic images and provide fine-grained control of image content. Diffusion models have quickly surpassed GANs as the generative method of choice because they can train stably and produce high-quality images [13-16]. Conditional diffusion models with text prompts (e.g., Stable Diffusion) can produce photorealistic images for broad categories of concepts. However, text prompts tend to be too abstract to accurately constrain spatial arrangements of multiple objects within an image. To mitigate this, layout-conditioned generation has emerged [17]. By conditioning on a spatial layout (a collection of bounding boxes and class labels), the model acquires explicit control of the placement and identity of the objects (where and what objects appear).

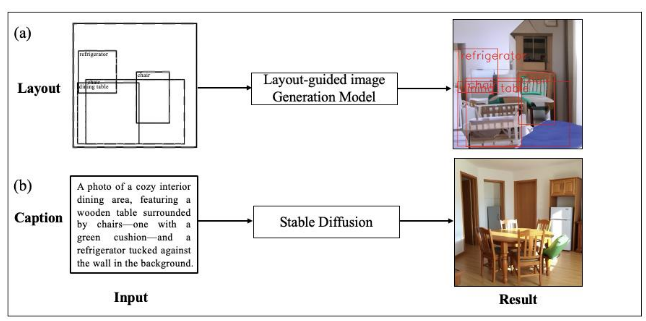

Fig. 1. Comparison of layout-guided vs. caption-guided image generation.

Figure 1(a) shows a layout-guided pipeline: a set of object bounding-box layouts (left) is fed into the layout-conditioned diffusion model, which produces an image (right) in which each object appears precisely at its specified spatial location. Because the generator strictly respects the input boxes, the resulting image–layout pairs can be used directly as labeled training data for object detectors. Figure 1(b) illustrates a caption-guided baseline: a natural-language description (left) is provided to

Stable Diffusion, yielding an image (right) that contains only the objects mentioned in the text, but without any control over their positions. In this case, the spatial arrangement is unconstrained, so the output cannot serve as a direct substitute for bounding-box supervision.

Early layout-to-image approaches with GANs (e.g., LostGAN, Layout2Im) established the possibility of generating images from object layouts but were often riddled with artifacts and low fidelity [18], [19]. The recent LayoutDiffusion model applies the power of diffusion to this task, with state-of-the-art image fidelity and spatial control [20]. LayoutDiffusion integrates several key modules that are tailored to the object detection task – a Layout Fusion Module and an Object-aware Cross Attention – to better combine layout features and position every object accurately and naturally. LayoutDiffusion surpasses previous GAN-based methods on standard benchmarks with FID gains of 46% on the COCO-Stuff dataset and substantially better content alignment scores. This progress makes layout-to-image diffusion generation an attractive method for generating varied and label-consistent synthetic training data.

In this work, we propose a new layout-to-image driven large-scale data augmentation for the object detection task. The LayoutDiffusion model is employed as the core generative model to synthesize novel images conditioned on bounding box layouts from existing object detection datasets. This allows us to automatically create novel training examples that adhere to the spatial distributions and object compositions of the original, but with novel visual instantiations. More importantly, the synthesized image includes ground-truth annotations "for free" – the bounding boxes provided to the model act as the labels for the synthesized image. This method solves the labeling bottleneck directly: with a diffusion model trained (or fine-tuned) on the domain, an infinite number of labeled images can be synthesized without extra human labeling. Our proposed approach not only expands data variety, but also adds new object appearances, backgrounds, and general scene compositions that are not available in the original dataset.

To evaluate the effectiveness of our proposed data augmentation framework, we use general object detection benchmarks (i.e., COCO and PASCAL VOC) to prove that our augmentation pipeline can achieve consistent improvements for contemporary detectors under the common training setup [21]. This emphasis differs from previous work that often applied generative augmentation mostly to the few-shot scenarios or specialized domains [14]. We can conclude that under an ordinary, fully supervised setup with tens of thousands of images, synthetic data can still enhance detection accuracy.

The contributions of this work can be summarized as:

- We introduce a new object detection data augmentation pipeline using layout-to-image diffusion models. Realistic images with embedded annotations can be generated by conditioning on bounding box layouts, without additional labeling efforts. To the best of our knowledge, this is the first work that integrates the layout-to-image framework for object detection data augmentation.
- Extensive experimental results prove that we can use synthesized images (along with their automatically generated annotations) in typical detector training to improve the accuracy, mean Average Precision (mAP), and enhance the robustness of object detection models.
- We thoroughly examine the qualitative properties of the synthetic data and the effects on training. Several novel insights have been found.

The rest of the paper is structured as follows. We review existing data augmentation methods for detection, diffusion-based generation, and layout-to-image synthesis in Related Work. In Methodology, we introduce the LayoutDiffusion model and our pipeline for data augmentation in detail, including the mathematical formulation of layout-conditioned diffusion and our task-specific adaptations. Quantitative evaluations on detection benchmarks and diversity vs. quantity are presented in the Experiments. We conclude with insights for generative data augmentation on object detection.

II. Related Works

A. Data Augmentation for Object Detection

Data augmentation has been used to increase the effective size and variability of training datasets for object detection [9]. Simple model-free augmentations involve geometric and photometric transformations applied to existing images, such as random flipping, rotation, scaling, color jitter, cropping, and mosaic combinations of images. These techniques preserve the integrity of the image and its annotations (bounding boxes are adjusted accordingly), thus maintaining semantic consistency. The augmented image still depicts the same scene and objects as the original. These methods improve the model's invariance to object position and appearance. However, they can only produce limited diversity [12], [20]. The synthetic samples remain close variants of original images, and no truly novel object instances or backgrounds are introduced.

To increase diversity, researchers explored instance-level augmentation and content mixing. One such method is copy-paste augmentation, which involves cutting out an object from one image and pasting it into the context of another image [22], [23]. By synthesizing objects into new scenes, copy-paste augmentation can create unusual combinations that do not exist in the dataset, potentially enabling detectors to handle a broader range of scenarios.

Another research direction is generative data augmentation, using learned models to synthesize images. Unlike geometric transforms or copy-paste, generative methods can create entirely new imagery, thus offering potentially unlimited diversity. Early attempts used classical generative adversarial networks (GANs) to produce additional training images. [13], [24], [12]. For classification tasks, this has shown success (e.g., GAN-generated images improving image classifiers). However, extending this to object detection is non-trivial because it is very challenging to produce corresponding bounding box labels for the synthetic images. Some works sidestep this by generating images and then applying an existing detector to annotate them. However, this depends on a high-quality pre-trained detector and can introduce labelling errors [9]. Ideally, the generative process should be controllable so that we know the location and category of each object a priori.

B. Diffusion Models and Layout-to-Image Generation

Recently, diffusion models have spurred new augmentation strategies for object detection. Diffusion-based augmentation can leverage powerful text-to-image models (like Stable Diffusion) to generate images containing particular objects or scenes described by a prompt [10], [15]. However, without explicit spatial control, the resulting image's object placement is unpredictable, making it hard to produce precise bounding box annotations for training.

Layout-based generation was previously explored with GANs[17], [18], [19], [24]. Those methods established the concept of generating images from bounding box arrangements but were limited by GAN image quality. For example, LostGAN and its improved version, LostGAN-v2, could generate simple multi-object scenes, but often with blurred details and artifacts, especially for complex scenes with many objects. More recent solutions like OC-GAN and Grid2Im sought better object compositionality, yet still lag behind real image quality [25]. With diffusion models, the fidelity has greatly improved. The LayoutDiffusion model is a landmark in this area. It integrates the strengths of latent diffusion with layout conditioning, resulting in images that are significantly more photorealistic and correctly aligned to the input layout than prior work [20].

A recent study presented a detection-oriented diffusion-based augmentation pipeline [14]. They employ a controllable diffusion model with CLIP-guided generation to make sure every produced image includes proper "visual priors" (i.e., a coarse layout on what should go where). This allows them to produce simulated images and accurate bounding box annotations. However, their pipeline still relies on coarse visual priors (e.g., HED edges or segmentation masks) and CLIP-based post-hoc filtering to enforce layout fidelity, which can leave residual misalignments and label noise in the synthetic data.

III. Methodology

In this section, we present our strategy for enriching an object detection dataset with automatically generated images that naturally carry precise bounding-box labels. First, we introduce how object layouts are represented and integrated into a layout-to-image diffusion generator. Then, the complete augmentation pipeline is presented, covering sample generation, quality, layout consistency filtering, and adopting synthetic examples into detector training.

A. Diffusion-based Layout-to-Image Generation Model

Layout Representation. Each input layout is a set of m object descriptors:

$$\mathcal{L} = \{(\boldsymbol{b_i}, c_i)\}_{i=1}^{m} \tag{1}$$

where $\boldsymbol{b_i} = (x_{i0}, y_{i0}, x_{i1}, y_{i1})$ are normalized coordinates of the ith bounding box and c_i is its class label. To form a fixed-length token sequence:

1. embed box coordinates: $\boldsymbol{b_i}$ is projected by a learnable matrix $\boldsymbol{W_b} \in R^{4\times d}$ into a d-dimensional vector.
2. embed class labels: c_i is converted via an embedding lookup $\boldsymbol{W_c} \in R^{C\times d}$.
3. sum embeddings: The box and label embeddings are summed into an object token $\boldsymbol{o_i} \in R^{d}$.

A global token $\boldsymbol{o_0}$ (covering the entire image) is prepended with padding tokens $\boldsymbol{o_p}$ to reach a constant length K. This yields a token sequence:

$$[\boldsymbol{o_0}, \boldsymbol{o_1}, \dots, \boldsymbol{o_m}, \boldsymbol{o_p}, \dots, \boldsymbol{o_k}] \tag{2}$$

Layout Fusion Module. To capture relationships among objects (e.g. overlap, contextual co-occurrence), we pass these K tokens through a small transformer encoder. Multi-head self-attention in this module allows each token to exchange information with all others, producing a fused layout representation:

$$\boldsymbol{L'} = \text{TransformerEncoder}([\boldsymbol{o_0}, \dots, \boldsymbol{o_{K-1}}]) \tag{3}$$

where $\boldsymbol{L'} \in R^{K\times d}$.

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where $\boldsymbol{L'} \in R^{K\times d}$.

Image-Layout Fusion in the Denoiser. The core image generator is a denoising U-Net trained under the diffusion framework. At each denoising timestep t, the network receives a noisy image (or latent) x_t and must predict the added noise ϵ.

This process is conditioned on the fused layout $\boldsymbol{L'}$ in two ways:

1. Global conditioning: After each down- or up-sampling block in the U-Net, a learned linear projection of the global token L'_0 (the first row) is added to the feature maps. This biases the entire image toward the overall scene context.
2. Object-aware cross-attention: Within selected U-Net layers, reinterpret the feature map is reinterpreted as a set of spatial queries $\{\boldsymbol{f_p}\}$, each corresponding to one image patch p. These queries are augmented with positional embeddings of the patch coordinates, ensuring they occupy the same spatial frame as the layout tokens. Cross-attention is then performed where queries $\boldsymbol{f_p}$ attend to keys/values drawn from all $\boldsymbol{L'}$ tokens. This allows the network to inject object-specific details precisely at the intended locations.

Formally, for a given patch feature $\boldsymbol{f_p} \in R^{d}$,

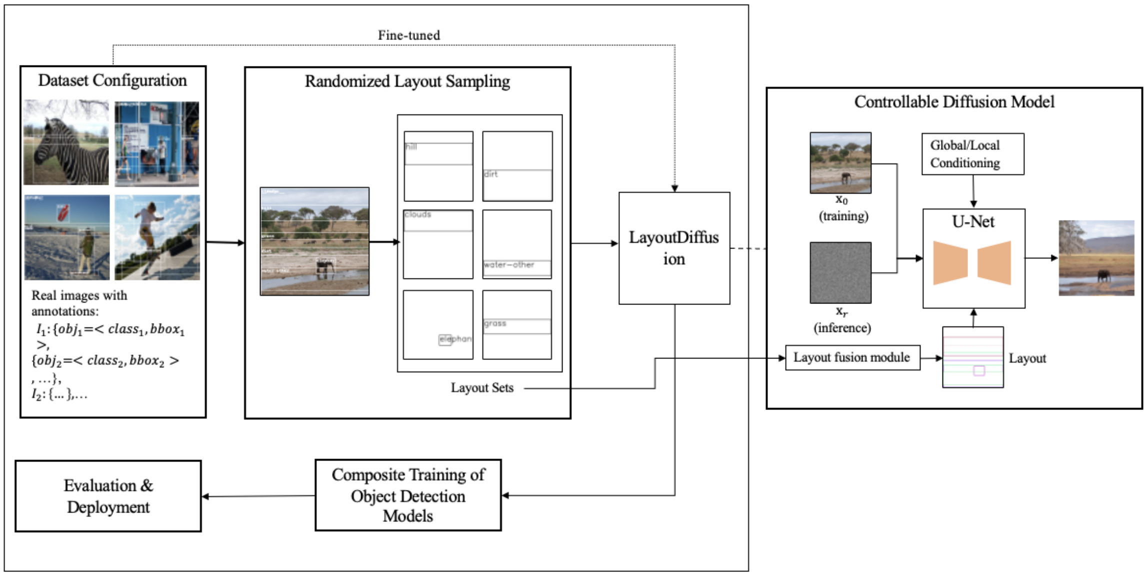

Fig. 2. Overview of the proposed layout-guided augmentation pipeline. Starting from real images with bounding-box annotations, we randomly sample or perturb layouts and feed them—via a transformer-based fusion module and global/local conditioning—into a fine-tuned diffusion U-Net to generate spatially accurate images. Synthetic outputs then undergo quality and consistency filtering before being combined with the original data for training and evaluation of object detectors.

$$\text{Attention}(\boldsymbol{f_p}, \boldsymbol{L'}, \boldsymbol{L'}) = \text{softmax}\left(\left(\boldsymbol{W_{qf_p}}\right)(\boldsymbol{W_{kL'}})^\top/\sqrt{d}\right)\boldsymbol{W_{vL'}} \quad (5)$$

where $\boldsymbol{W_q}, \boldsymbol{W_k}, \boldsymbol{W_v}$ are learned projections. The result is added back to $\boldsymbol{f_p}$, enriching it with layout-guided content.

Training and sampling. model is trained with the standard denoising objective:

$$\mathbb{E}_{x_0,\mathcal{L},t,\epsilon}[\| \epsilon - \epsilon_\theta(x_t, t, \mathcal{L}) \|^2] \quad (6)$$

where x_t is a noisy version of a real image x_0 matching layout $\mathcal{L}$. During sampling, classifier-free guidance are applied by interpolating between the conditioned and unconditioned noise predictions to balance fidelity to the layout and image quality.

B. *Synthetic Data Generation Pipeline*

Let

$$\mathcal{D}_{\text{real}} = \{(I^{(j)}, \mathcal{B}^{(j)})\}_{j=1}^{N} \quad (7)$$

$$\mathcal{D}_{\text{syn}} = \{(\tilde{I}^{(k)}, \tilde{\mathcal{B}}^{(k)})\}_{k=1}^{M} \quad (8)$$

denote the real and synthetic datasets, where

$$\mathcal{B}^{((j))} = \{b_i^{((j))}\}_{i=1}^{(n_j)}, b_i^{(j)} = (x_{i0}, y_{i0}, x_{i1}, y_{i1}, c_i) \quad (9)$$

is the set of n_j bounding boxes and class labels in image $I^{(j)}$.

Our pipeline proceeds in three stages:

Layout Selection and Generation. We begin by sampling real object layouts from the existing dataset and, where desired, introduce diversity by slightly perturbing bounding-box positions and, in some cases, swapping object classes with semantically related categories. This yields a mix of faithful reproductions and intentionally varied layouts for augmentation.

Diffusion-based Image Synthesis. Each selected (or perturbed) layout is provided to our fine-tuned diffusion model, which converts structured noise into a fully rendered image. Through successive denoising steps, the model generates realistic scenes in which each object faithfully occupies its prescribed box.

Augmented Training. The selected synthetic images and their labels are combined with the real training set. We typically shuffle and mix them so that each training batch contains a proportion of real and synthetic data. We define augmentation ratio $\alpha = M/N$. The detector is trained on shuffled batches containing both real and synthetic images, with standard augmentations (e.g., flips, color jitter) applied uniformly. We experiment with different α to find an optimal balance.

IV. Experiments

We conduct experiments on two widely used object detection benchmarks, COCO and PASCAL VOC, to evaluate the effectiveness of the proposed layout-to-image augmentation framework. We first describe the experimental setup: datasets, generation settings, and detection models. Then we present the main results comparing baseline training vs. augmented training.

A. *Datasets and Setup*

COCO 2017. The MS COCO dataset is a large-scale detection benchmark with 80 object categories. We use the COCO 2017 train split (around 118k images) for training and the val split (5k images) for evaluation of mAP. For our augmentation, 20 classes are selected (to parallel VOC categories) to generate images. However, the detectors are always evaluated on all 80 classes. Most COCO images contain multiple objects; we use their full layouts for generation.

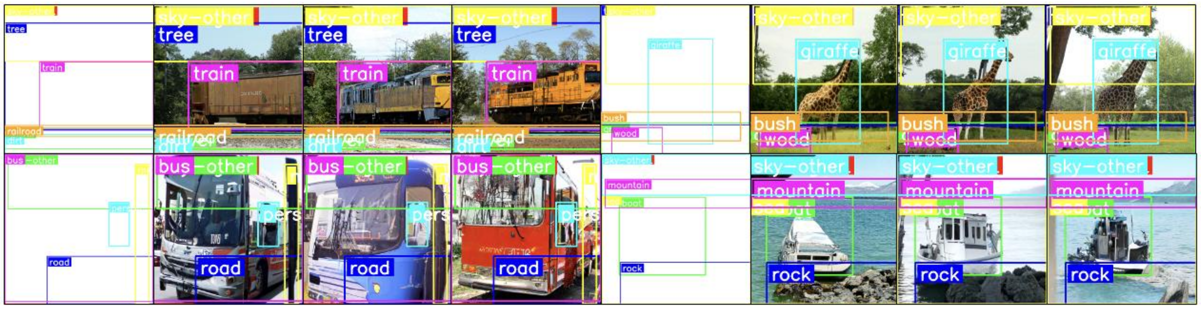

Fig. 3. Visualization of our results. Each row shows a different pair of input layouts: columns 1 (and 5) display the two layouts with bounding boxes and class labels, and columns 2–4 (and 6–8) show three generated image samples for that layout, each annotated with same boxes and labels.

PASCAL VOC. We use the VOC 2007+2012 combined dataset for training (~16.5k images with 20 classes) and evaluate on the VOC 2007 test (4952 images). VOC has fewer images and classes, making it a suitable testbed for augmentation. Many VOC images have only 1 or 2 objects, and the scenes are less diverse than COCO. We generate synthetic images for VOC in two modes: (a) using VOC's own layouts (which are simple) and (b) using some COCO layouts filtered to only include VOC classes (to test generating slightly more complex scenes for VOC).

Generation Details. We use a LayoutDiffusion model pre-trained on COCO-Stuff (which covers COCO's 80 classes plus background stuff) at 256×256 resolution. For VOC, which has similar classes (a subset of COCO's, like person, car, dog, etc.), this model is still applicable.

We generate 30k images for COCO and 5k for VOC as candidate pools. These numbers correspond to roughly 25% of the COCO train and 30% of the VOC train in terms of additional data. We tried larger pools (up to 2x of the original dataset size) but found diminishing returns. The guidance scale w of the diffusion model is set to 4.0 for most samples; we also generate ~20% of images with w=2.0 (more diverse, slightly less faithful) to increase variety. The diffusion model inference uses 50 denoising steps (DDIM sampler) per image to balance quality and runtime.

B. Detectors

YOLOv8. It is a one-stage anchor-free detector known for its speed and accuracy on COCO. We use the yolov8m variant (medium model) pre-trained on COCO as a starting point, then fine-tune on each dataset (COCO or VOC) with or without augmentation. YOLOv8m achieves about 50.2% mAP on COCO val when trained on COCO normally. For VOC, it reaches ~82% mAP@0.5 (since VOC mAP is typically reported at 0.5 IoU threshold).

Faster R-CNN. It is a classic two-stage detector with a Region Proposal Network and a ResNet-50 backbone. We train it from scratch on each dataset using Detectron2 with standard hyperparameters (e.g., 1× schedule for COCO). This model gives a point of comparison to see if augmentation benefits both one-stage and two-stage frameworks similarly.

All models are trained on the augmented dataset for the same number of epochs as the baseline. We ensure each training sees the same number of real image iterations – for fairness, if we double the data, we also double training iterations so that the model still sees real images the same number of times (plus extra synthetic).

C. Metrics

Mean Average Precision (mAP). For COCO, we report the standard mAP at IoU 0.5:0.95. For VOC, we report the conventional AP@0.5 (since that is the usual VOC metric).

Fréchet Inception Distance (FID). We calculate FID between generated images and real images for each dataset to assess the image realism.

D. Result

We first evaluate the impact of the proposed synthetic augmentation solution on two standard detection benchmarks. Table I reports results on COCO val2017 using YOLOv8-m and Faster R-CNN, both trained on 10,000 real images without and with 10,000 additional synthetic samples ($\alpha = 1$). The accuracy of the YOLOv8-m model increases by 2.46%, and Faster R-CNN increases by 3.12%. Table II shows analogous experiments on PASCAL VOC2007 (mAP@0.5), where we add 10,000 synthetic images (α =1). Here, YOLOv8-m increases by 2.75 % and Faster R-CNN increases by 3.14%. These gains across one-stage and two-stage detectors underscore the effectiveness and generality of our layout-guided augmentation.

TABLE I. RESULT ON COCO

Model	Baseline mAP (0.5:0.95)	Augmented mAP (0.5:0.95)	Δ (pp)	Δ (%)
YOLOv8-m	50.2	51.43	+1.23	+2.46%
Faster R-CNN	38.5	39.7	+1.20	+3.12%

TABLE II. RESULT ON PASCAL VOC

Model	Baseline mAP@0.5 (%)	Augmented mAP@0.5 (%)	Δ (pp)	Δ (%)
YOLOv8-m	88.5	90.93	2.43	+2.75%
Faster R-CNN	73.3	75.60	2.30	+3.14%

We then conduct a series of controlled studies on the COCO dataset to study how synthetic augmentation affects detection performance, data diversity, and image quality. All experiments use YOLOv8m trained on a fixed set of 10,000 real images, with synthetic samples generated by the layout-to-image diffusion model. The four analyses below reveal (1) how increasing synthetic volume yields diminishing returns, (2) the effect of varying the real and synthetic ratio, (3) diversity saturation across large synthetic pools, and (4) image quality trends via FID.

Impact of Synthetic Volume on Detection Accuracy. We fix the real training set at 10,000 COCO images and add synthetic images in five volume increments: 1 % (100), 5 % (500), 10 % (1,000), 50 % (5,000), and 100 % (10,000). The results in Table III illustrate a clear diminishing returns trend: initial volumes of synthetic samples contribute substantial improvements, but marginal gains shrink as more data is added. In practice, adding synthetic images equivalent to 50 % to 100 % of the real dataset captures nearly all achievable benefit while controlling generation overhead.

TABLE III. MAP GAIN (IN PERCENTAGE POINTS) ON COCO VAL AS A FUNCTION OF SYNTHETIC DATA VOLUME.

Synthetic%	**+ 1%**	**+ 5%**	**+ 10%**	**+ 50%**	**+ 100%**
Total Images	10100	10500	11000	15000	20000
mAP (pp)	+ 0.05	+ 0.12	+ 0. 21	+ 0.77	+ 1.23

Effect of Real and Synthetic Training Ratio. We fix the total number of training samples as 10000, and vary the proportion of real versus synthetic images to study its impact on detection accuracy. Using YOLOv8m on COCO, we consider the following ratios of real and synthetic images: 10 : 0, 9 : 1, 5 : 5, 1 : 9, and 0 : 10. As shown in Table IV, we can find that although synthetic examples provide a valuable supplement, they cannot replace genuine annotations. This finding highlights the importance of maintaining a healthy majority of real data.

TABLE IV. MAP ON COCO UNDER DIFFERENT REAL : SYNTHETIC MIXTURES

Real : Synthetic	**10:0**	**8:2**	**5:5**	**2:8**	**0:10**
mAP	50.20	51.03 (+0.83)	51.43 (+1.43)	47.69 (-2.51)	41.67 (-8.53)

Diversity Saturation in Large-Scale Generation. We assess how the **intra-set diversity** of synthetic images evolves with pool size. Using LPIPS as a perceptual-similarity measure (higher LPIPS → more diverse), we sample 1,000 pairs at each synthetic-data volume. Diversity drops noticeably beyond about 10k samples (from 0.437 to 0.402 at 20k), confirming that over-sampling from a fixed generative model leads to redundant outputs. This finding suggests that we should use different layout inputs to enhance the diversity of the generated dataset.

TABLE V. MAP ON COCO VAL UNDER DIFFERENT REAL : SYNTHETIC MIXTURES

Synthetic Pool	5000	10000	20000	30000
LPIPS ↑ (avg.)	0.430	0.437	0.402	0.385

FID and Image Quality Trends. Finally, we examine the Fréchet Inception Distance (FID) between synthetic and real COCO images at different generation scales. As shown in Table VI, FID steadily decreases when more images are generated. This pattern aligns with observations in GAN and diffusion models. Emphasizing the need to balance generation scale against computational cost and diminishing improvement.

TABLE VI. MAP ON COCO VAL UNDER DIFFERENT REAL : SYNTHETIC MIXTURES

Synthetic Pool	5000	10000	20000	30000
FID ↓	24.1	18.2	16.0	15.2

V. CONCLUSION AND FUTURE WORK

In this paper, we introduced a controllable diffusion-based augmentation strategy that synthesizes training images conditioned on bounding-box layouts. By combining real and generated samples, we achieved consistent mAP improvements on COCO and PASCAL VOC across multiple detector architectures, demonstrating the practicality of diffusion-based augmentation for general object detection.

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